

Semantic entropy pipeline

Farquhar et al. 2024 — bidirectional entailment clustering

PHASE 1 INPUTS

Context (problem description)

"Permutation bug: overlapping cycles produce wrong output"

Candidate 0

"Fix validation..."

Candidate 1

"Add type guard..."

Candidate 2

"Rewrite loop..."

Candidate 3

"Normalize inputs..."

N intent strings (from strategy proposals or LLM trajectory summaries)

PHASE 2 CONTEXT PREPENDING

Each intent gets the context string glued in front: f"{context} {intent}"

ctx + intent[0]

"Permutation bug... Fix validation in parse()"

ctx + intent[1]

"Permutation bug... Add input type guard..."

ctx + intent[2]

"Permutation bug... Rewrite the loop..."

ctx + intent[3]

"Permutation bug... Normalize inputs..."

Context gives DeBERTa the semantic frame — "modify constructor" and "fix disjointness check" become equivalent in context

PHASE 3 GREEDY BIDIRECTIONAL ENTAILMENT CLUSTERING

i = 0

No clusters exist → intent 0 becomes first cluster

C = { 0 }

= representative

i = 1

Compare intent 1 vs C representative (intent 0) — two DeBERTa passes

Forward: classify(ctx+intent[0], ctx+intent[1])

→ P(ent) = 0.87

0.87 > 0.5 AND 0.79 > 0.5 → MATCH

Backward: classify(ctx+intent[1], ctx+intent[0]) ← swapped

→ P(ent) = 0.79

Add to C

C = { 0 , 1 }

i = 2

Compare intent 2 vs C rep (intent 0)

Fwd: P(ent) = 0.12 Bwd: P(ent) = 0.09

0.12 < 0.5 → NO MATCH

No more clusters → new cluster

C = { 2 }

i = 3

Compare intent 3 vs C rep, then C rep

vs C : Fwd = 0.62, Bwd = 0.38

Fwd > 0.5 but Bwd < 0.5 → NO MATCH

No clusters matched → new

vs C : Fwd = 0.15, Bwd = 0.21

Both < 0.5 → NO MATCH

C = { 3 }

Result

K = 3 clusters from N = 4 intents

C = { 0, 1 }
size = 2

C = { 2 }
size = 1

C = { 3 }
size = 1

Total: 8 DeBERTa forward passes (2 per comparison × 4 comparisons). Complexity: O(N × K) pairs, 2 calls each.

PHASE 4 SEMANTIC ENTROPY COMPUTATION

$$H = -\sum p(c) \cdot \ln p(c)$$

where $p(c) = |\text{cluster } c| / N$ (fraction of intents in each cluster)

Step 1: Probabilities

$p(C_0) = 2/4 = 0.50$ $p(C_1) = 1/4 = 0.25$ $p(C_2) = 1/4 = 0.25$

Step 2: Entropy terms

$-0.50 \times \ln(0.50) = 0.347$ $-0.25 \times \ln(0.25) = 0.347$ $-0.25 \times \ln(0.25) = 0.347$

$$H = 0.347 + 0.347 + 0.347 = 1.040$$

PHASE 5 BRANCHING DECISION

$$\text{should_branch} = (H > \tau) \rightarrow (1.040 > 0.5) \rightarrow \text{True}$$

H > τ → BRANCH

Fork into K = 3 parallel trajectories
(one per cluster, using cluster representative)

H ≤ τ → DON'T BRANCH

Pick best candidate, continue single trajectory
(model is confident — no real disagreement)

Entropy range: $H = 0$ (all in one cluster, fully converged) to $H = \ln(K)$ (uniform across K clusters, maximum diversity)